

Total No. of Questions : 8]

SEAT No. :

PC2381

[Total No. of Pages : 2

[6354]-498

B.E. (Computer Engineering)
HIGH PERFORMANCE COMPUTING
(2019 Pattern) (Semester - VIII) (410250)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.

- Q1)** a) Explain it with the help of algorithm for all-to-all broadcast on hypercube. [6]
b) Describe One-to-All Broadcast and All-to-One Reduction on Rings. [6]
c) Explain how non-blocking communication is finalised. [5]

OR

- Q2)** a) Explain Prefix-Sum Operations on eight-node hypercube with help of diagram. [6]
b) Explain All-to-All Broadcast and Reduction on a Ring with help of algorithm. [6]
c) What is non-blocking communication? Explain isend, irecv methods. [5]

- Q3)** a) With help of example, explain Effect of Granularity on Performance. [6]
b) Describe a situation where fine-grained parallelism might be more appropriate than coarse-grained parallelism for improving scalability [6]
c) Describe Sources of Overhead in Parallel Programs and comments on how to avoid it. [5]

OR

- Q4)** a) What are the key factors that can limit the scalability of a parallel application, and how can they be addressed? [6]
b) Explain Isoefficiency Metric of Scalability and Describe a scenario where an isoefficiency value of 1 would indicate perfect scalability. [6]
c) Enlist performance Metrics for Parallel Systems and explain any two. [5]

P.T.O.

- Q5)** a) Explain the concept of parallel processing in CUDA architecture and how it differs from traditional CPU-based computing. [8]
- b) Explain the distinction between global memory and shared memory in CUDA. How are they used differently? [6]
- c) Describe how communication managed and synchronized in CUDA. [4]

OR

- Q6)** a) Explain CUDA memory model with help of CU DA memory hierarchy.[8]
- b) Explain processing flow of CUDA along with CUDA C functions used.[6]
- c) Enlist and explain applications of CUDA. [4]

- Q7)** a) Explain parallel Bubble sort with help of algorithm. [8]
- b) Explain Kuberbets framework with help of diagram. [6]
- c) Explain how Document classification works in Distributed Computing.[4]

OR

- Q8)** a) Explain parallel Merge sort with help of algorithm. [8]
- b) Explain the concept of container orchestration in Kubernetes. How does it simplify application deployment and management? [6]
- c) Elaborate use of AI/ML in Parallel Computing. [4]

